(i) All the elements of $\mathscr{P}(S)$ are events.

(ii) $\varnothing$ is an impossible event.

(iii) $\{1\},\{2\},\{3\},\{4\}$ are the simple events or elementary events.

(iv) $\{1, 2, 3, 4\}$ is a sure event or certain event.

Definition 12.8

Two events cannot occur simultaneously are mutually exclusive events. $A_1, A_2, A_3, ..., A_k$ are **mutually exclusive** means that, $A_i \cap A_j = \varnothing$, for $i \neq j$.

Definition 12.9

Two events are **mutually inclusive** when they can both occur simultaneously.

$A_1, A_2, A_3, ..., A_k$ are mutually inclusive means that, $A_i \cap A_j \neq \varnothing$, for $i \neq j$

Illustration 12.3

When we roll a die, the sample space $S = \{1,2,3,4,5,6\}$.

(i) Since $\{1, 3\} \cap \{2, 4, 5, 6\} = \varnothing$, the events $\{1,3\}$ and $\{2, 4,5,6\}$ are mutually exclusive events.

(ii) The events $\{1,6,\}, \{2,3,5\}$ are mutually exclusive.

(iii) The events $\{2,3,5\}, \{5,6\}$ are mutually inclusive, since $\{2, 3, 5\} \cap \{5, 6\} = \{5\} \neq \varnothing$

Definition 12.10

$A_1, A_2, A_3, ..., A_k$ are called **exhaustive events** if, $A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_k = S$

Definition 12.11

$A_1, A_2, A_3, ..., A_k$ are called **mutually exclusive and exhaustive events** if,

(i) $A_i \cap A_j = \varnothing$, for $i \neq j$ (ii) $A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_k = S$

Illustration 12.4

When a die is rolled, sample space $S = \{1,2,3,4,5,6\}$.

Some of the events are $\{2,3\}, \{1,3,5\}, \{4,6\}, \{6\}$ and $\{1,5\}$.

(i) Since $\{2, 3\} \cup \{1, 3, 5\} \cup \{4, 6\} = \{1, 2, 3, 4, 5, 6\} = S$ (sample space), the events $\{2,3\}, \{1,3,5\}, \{4,6\}$ are exhaustive events.

(ii) Similarly $\{2,3\}, \{4,6\}$ and $\{1,5\}$ are also exhaustive events.

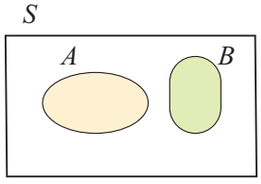
(iii) $\{1,3,5\}, \{4,6\}, \{6\}$ and $\{1,5\}$ are not exhaustive events.

(Since $\{1, 3, 5\} \cup \{4, 6\} \cup \{6\} \cup \{1, 5\} \neq S$)

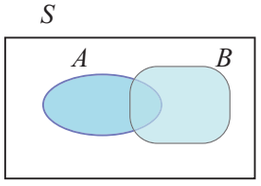
(iv) $\{2,3\}, \{4,6\}$, and $\{1,5\}$ are mutually exclusive and exhaustive events, since

$\{2,3\} \cap \{4,6\} = \varnothing, \{2,3\} \cap \{1,5\} = \varnothing, \{4,6\} \cap \{1,5\} = \varnothing$ and $\{2, 3\} \cup \{4, 6\} \cup \{1, 5\} = S$

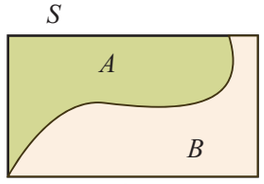
Types of events associated with sample space are easy to visualize in terms of Venn diagrams, as illustrated below.



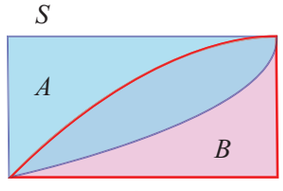
A and B are Mutually exclusive



A and B are Mutually inclusive



A and B are Mutually exclusive and exhaustive



A and B are Mutually inclusive and exhaustive

Definition 12.12

The events having the same chance of occurrences are called **equally likely** events.

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